

Clifford Algebras and the Kepler Problem in Theoretical Physics

Seminar Methodology, Schedule, and Foundational Syllabus

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1 Overview

This seminar investigates the Kepler problem across four physical domains unified under the invariant language of Clifford geometric algebras. The progression spans classical central force dynamics in $\mathcal{Cl}(3, 0)$, spinor regularization, relativistic motion and gauge gravity in the spacetime algebra $\mathcal{Cl}(1, 3)$, and the relativistic quantum field description of the hydrogen atom via the Dirac–Hestenes equation.

The colloquium operates as a working research seminar. Participants engage as collaborating researchers responsible for analytical rigour, formal mathematical derivations, and critical evaluation of physical foundations. The program comprises 18 weekly sessions of 90 minutes. Weeks 1 through 3 establish the algebraic foundation, while Weeks 4 through 18 are conducted by senior–junior tandems.

2 Methodology

2.1 Working Roles

The seminar avoids passive lecture formats. Each session relies on four distinct responsibilities.

Referat (Lead Exposition) Delivered jointly by the assigned tandem. The presentation focuses on direct, blackboard derivations of primary equations. Conceptual assumptions and algebraic transitions take precedence over projected slides.

Korreferat (Respondent Critique) An assigned participant presents a critical commentary immediately following the Referat. The respondent examines implicit hypotheses, compares the results with traditional coordinate or matrix formalisms, identifies potential pathologies, and formulates targeted questions for discussion.

Thesenpapier (Position Paper) A concise memorandum circulated by the presenting tandem at least five days prior to the session. It contains formal definitions, key algebraic identity statements, and explicit theses concerning physical interpretation. Participants study this document in advance.

Sitzungsprotokoll (Session Record) A rotating participant documents the derivation steps, the resolution of theoretical debates, and open problems. The finalized protocol is archived in the shared repository within seven days following the session.

2.2 Tandem Collaboration

Weeks 4 through 18 pair an advanced participant (senior) with an earlier-stage student (junior). The senior ensures mathematical coherence and contextualizes advanced algebraic techniques. The junior works directly through the derivation machinery, receiving structured guidance in multilinear structures and relativistic dynamics. Responsibilities for blackboard presentation are shared equally between both speakers.

2.3 Session Structure

Preparatory meetings with the seminar director occur two weeks and five days prior to each session. Every attendee must submit at least one written question on the Thesenpapier 24 hours before meeting. The 90-minute session adheres to the schedule in table 1.

Table 1: Allocation of time per 90-minute seminar session.

Phase	Duration	Content	Responsible
I	40 min	Live blackboard derivation (Referat)	Presenting tandem
II	10 min	Critical response and opening theses (Korreferat)	Respondent
III	35 min	Plenary debate, boundary cases, interpretation	All participants
IV	5 min	Synthesis of open questions and preview	Director

3 Schedule and Syllabus

Primary reference works are designated as follows: **VR** indicates Vaz and da Rocha, *An Introduction to Clifford Algebras and Spinors* [1]; **DL** indicates Doran and Lasenby, *Geometric Algebra for Physicists* [2]; **HS** indicates Hestenes and Sobczyk, *Clifford Algebra to Geometric Calculus* [3]. Specific chapters and research papers are cited directly in table 2.

Table 2: Complete 18-week seminar schedule and syllabus.

Wk	Topic	Mathematical and Physical Content	Literature
<i>Part A: Foundations and Geometric Analysis</i>			
1	Dual spaces, multilinear forms, and quadratic spaces	Dual linear space V^* , dual pairing $\langle \cdot, \cdot \rangle$, tensor algebra $T(V)$, symmetric bilinear forms, quadratic forms $q(v)$, polarization, non-degeneracy, and signature (p, q) over $\mathbb{R}$.	
2	Universal Clifford construction and involutions	Ideal $\mathcal{I}_q = \langle v \otimes v - q(v)1 \rangle$, quotient algebra $\mathcal{C}\ell(V, q)$, universal property, $\mathbb{Z}_2$ -grading, grade involution $\hat{u}$, reversion $\tilde{u}$, Clifford conjugation $\bar{u}$, geometric product $uv = u \cdot v + u \wedge v$.	
3	Multivector structures, Lie algebras, and Spin groups	Module isomorphism $\mathcal{C}\ell(V, q) \cong \bigwedge V$, blade projections, bivector Lie algebra $\mathcal{C}\ell^2(V, q) \cong \mathfrak{so}(p, q)$, Clifford and Lipschitz groups, $\text{Pin}(p, q)$, $\text{Spin}(p, q)$, and rotor transformations $v \mapsto Rv\tilde{R}$.	

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Wk	Topic	Mathematical and Physical Content	Literature
4	Clifford analysis and monogenic fields	Vector derivative ∇ , Cauchy–Riemann generalized operator, monogenic multivector fields $\nabla\Phi = 0$, Cauchy kernel, Borel–Pompeiu integral formula, and coordinate-free multipole expansions of extended Keplerian sources.	
<i>Part B: Classical Kepler Dynamics in $\mathcal{Cl}(3,0)$</i>			
5	Central forces and orbital angular momentum (Tandem 1)	Equation of motion $\dot{p} = -(k/r^3)r$, orbital angular momentum bivector $L = r \wedge p$, conservation proof $\dot{L} = 0$, planar confinement $r \wedge L = 0$, coordinate-free duality via volume pseudoscalar I_3 .	
6	Laplace–Runge–Lenz vector and conics (Tandem 2)	Multivector construction $A = \frac{1}{mk}pL - \hat{r}$, algebraic proof of $\dot{A} = 0$ without coordinate frames, orbit derivation via scalar contraction $r \cdot A$, and invariant classification of conics.	
<i>Part C: Regularization, Harmonic Analysis, and Hidden Symmetries</i>			
7	Kustaanheimo–Stiefel transformation (Tandem 3)	Regularization of collision singularities at $r \rightarrow 0$, spinor mapping $x = u\sigma_3\tilde{u}$ with $u \in \mathcal{Cl}^+(3,0) \cong \mathbb{H}$, Sundman time rescaling $dt = r ds$, and $U(1)$ gauge freedom.	
8	Four-dimensional oscillator and $SO(4)$ (Tandem 4)	Linearization to $u'' + \frac{ E }{2}u = 0$, four-dimensional isotropic harmonic oscillator, realization of $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ dynamical symmetry via L and A , bound-state degeneracy.	
9	Clifford–Fourier analysis and Fock projection (Tandem 5)	Clifford–Fourier transform with pseudoscalar kernel $e^{-I_3 k \cdot x}$, multivector uncertainty relations, Fock stereographic projection of the Coulomb problem onto S^3 , and bound states as spherical monogenics.	
<i>Part D: Relativistic Dynamics in Spacetime Algebra $\mathcal{Cl}(1,3)$</i>			
10	Spacetime algebra and observer split (Tandem 6)	Dirac generators $\{\gamma_\mu\}$, signature $(+, -, -, -)$, observer split via γ_0 , relative spatial bivectors $\sigma_k = \gamma_k \gamma_0$, proper Lorentz rotors in $\text{Spin}^+(1,3)$, rotor kinematics $\dot{R} = \frac{1}{2}\Omega R$, Thomas precession.	

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Wk	Topic	Mathematical and Physical Content	Literature
11	Relativistic electrodynamics and Lorentz force (Tandem 7)	Spacetime vector derivative $\nabla = \gamma^\mu \partial_\mu$, Faraday bivector $F = E + I_4 B$, Maxwell system $\nabla F = J$, relativistic equation of motion $\dot{p} = eF \cdot u$, proper-time rotor integration.	
12	Relativistic central forces and perihelion advance (Tandem 8)	Relativistic central potentials, breakdown of LRL vector conservation, energy-dependent dynamical corrections, and calculation of special-relativistic perihelion advance.	
13	Gauge Theory Gravity and planetary orbits (Tandem 9)	Position gauge field $\bar{h}(a)$ and rotation gauge field $\bar{\Omega}(a)$ on Minkowski background, spherically symmetric solutions, geodesic rotor trajectories, and general-relativistic orbital precession.	
14	Geometric thermodynamics in spacetime algebra (Tandem 10)	Conserved particle flux $N = nu$, stress-energy operator $T(a)$, entropy current $S = su$, invariant thermal 1-vector $\beta = u/(k_B T)$, Killing equilibrium $\nabla \wedge \beta = 0$, and Tolman–Ehrenfest thermal gradients in static gravitational fields.	
<i>Part E: Quantum Kepler Problem and Dirac–Hestenes Theory</i>			
15	Real Pauli algebra and spatial spin (Tandem 11)	Spatial algebra $\mathcal{Cl}(3, 0)$ without matrix representations, unit spatial vectors as spin observables, Pauli multivector wave equation, spin-orbit coupling from the geometric product.	
16	Dirac–Hestenes field equation (Tandem 12)	Spacetime algebra representation of the Dirac equation, even multivector field $\Psi \in \mathcal{Cl}^+(1, 3)$, spin plane $\gamma_2 \gamma_1$, polar form $\Psi = \rho^{1/2} e^{I_4 \beta/2} R$, probability current $J = \Psi \gamma_0 \tilde{\Psi}$, spin bivector $S = \frac{1}{2} \hbar \Psi \gamma_3 \tilde{\Psi}$.	
17	Exact solution of the relativistic hydrogen atom (Tandem 13)	Separation of variables in spherical coordinates within $\mathcal{Cl}^+(1, 3)$, rotor spherical harmonics, radial differential equations in the even sub-algebra, derivation of the Sommerfeld–Dirac energy spectrum.	

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Wk	Topic	Mathematical and Physical Content	Literature
18	Fine structure, Darwin term, and zitterbewegung (Tandem 14)	Non-relativistic expansion of the Dirac–Hestenes equation, geometric origin of the Darwin correction, rotor phase dynamics, lightlike field circulation, field-theoretic interpretation of zitterbewegung.	

4 Tandem Assignments

The assignments for the tandem-led sessions are presented in table 3.

Table 3: Register of tandem assignments and respondents for Weeks 5 through 18.

Wk	Topic	Senior	Junior	Respondent
5	Central forces and bivectors			
6	LRL vector and conics			
7	KS transformation			
8	4D oscillator and $SO(4)$			
9	Clifford–Fourier and Fock projection			
10	Spacetime algebra			
11	Electrodynamics in STA			
12	Relativistic central forces			
13	Gauge Theory Gravity			
14	Geometric thermodynamics			
15	Pauli algebra and spin			
16	Dirac–Hestenes equation			
17	Relativistic hydrogen			
18	Fine structure and zitterbewegung			

References

- [1] Jayme Vaz and Roldão da Rocha. *Introduction to Clifford Algebras and Spinors*. Oxford: Oxford university press, 2016. 128 pp. ISBN: 978-0-19-878292-6.
- [2] Chris Doran and Anthony Lasenby. *Geometric Algebra for Physicists*. Cambridge: Cambridge University Press, 2013. 578 pp. ISBN: 978-0-521-48022-2. DOI: [10.1017/CB09780511807497](https://doi.org/10.1017/CB09780511807497).
- [3] David Hestenes and Garret Sobczyk. *Clifford Algebra to Geometric Calculus*. Dordrecht: Springer Netherlands, 1984. ISBN: 978-90-277-2561-5 978-94-009-6292-7. DOI: [10.1007/978-94-009-6292-7](https://doi.org/10.1007/978-94-009-6292-7). URL: <http://link.springer.com/10.1007/978-94-009-6292-7>.